Solving the model by hand

Assume by now you have done algebra to get equations (9) and (10) in the Presentation, p. 12:

$$0 = -\pi_t + \alpha_1 \pi_{t-1} + \alpha_2 \lambda_t + \alpha_3 z_t \quad (9)$$

$$0 = E_t\left[-\lambda_t + \alpha_4 \pi_t + \alpha_5 \lambda_{t+1} + \alpha_6 z_{t+1}\right] \quad (10)$$

The proposed policy functions are:

$$\pi_t = \eta_{\pi\pi} \pi_{t-1} + \eta_{\pi z} z_t \quad (a)$$

$$\lambda_t = \eta_{\lambda\pi} \pi_{t-1} + \eta_{\lambda z} z_t \quad (b)$$

$\longrightarrow$ Substitute into (9) and (10):

Wherever π_t appears substitute (a);
wherever λ_t appears substitute (b)

$\Rightarrow$

(9): $$0 = \underbrace{-\eta_{\pi\pi} \pi_{t-1} - \eta_{\pi z} z_t}_{-\pi_t} + \alpha_1 \pi_{t-1} + \underbrace{\alpha_2 \eta_{\lambda\pi} \pi_{t-1} + \alpha_2 \eta_{\lambda z} z_t}_{\alpha_2 \lambda_t}$$

$$+ \alpha_3 z_t$$

Factorize k_{t-1}, z_t

(9): $$\boxed{0 = (-\eta_{kk} + \alpha_1 + \alpha_2 \eta_{xk}) k_{t-1} + (-\eta_{kz} + \alpha_2 \eta_{xz} + \alpha_3) z_t}$$

Now (10):

$$0 = E_t\Big[\underbrace{-\eta_{\lambda k} k_{t-1} - \eta_{\lambda z} z_t}_{-\lambda_t} + \underbrace{\alpha_4 \eta_{kk} k_{t-1} + \alpha_4 \eta_{kz} z_t}_{\alpha_4 k_t} + \underbrace{\alpha_5 \eta_{\lambda k} k_t + \alpha_5 \eta_{\lambda z} z_{t+1}}_{\alpha_5 \lambda_{t+1}\text{, need to substitute } k_t\text{!}} + \alpha_6 z_{t+1} \Big]$$

$$0 = -\eta_{\lambda k} k_{t-1} - \eta_{\lambda z} z_t + \alpha_4 \eta_{kk} k_{t-1} + \alpha_4 \eta_{kz} z_t + \boxed{\alpha_5 \eta_{\lambda z} E_t(\rho z_t + \varepsilon_{t+1})}$$

$$+ \boxed{\alpha_6 E_t(\varphi z_t + \varepsilon_{t+1})}$$
$$+ \alpha_5 \eta_{xk}\eta_{kk} k_{t-1}$$
$$+ \alpha_5 \eta_{xk}\eta_{kz} z_t$$

(10):

$$\boxed{\begin{aligned} 0 = {} & k_{t-1}\left(-\eta_{xk} + \alpha_4 \eta_{kk} + \alpha_5 \eta_{xk}\eta_{kk}\right) \\ & + z_t\Big(-\eta_{xz} + \alpha_4 \eta_{kz} \\ & \qquad \boxed{+ \alpha_5 \eta_{xz}\varphi + \alpha_6 \varphi} \\ & \qquad + \alpha_5 \eta_{xk}\eta_{kz}\Big) \end{aligned}}$$

where we used $E_t(\varphi z_t + \varepsilon_{t+1}) = \varphi z_t$

Next: These 2 equations must hold simultaneously for any possible value of (k_{t-1}, z_t)

This requires that each coefficient (the sums in parentheses) must be 0.

Let us work with the coefficient for k_{t-1}.

There is one in (9), and one in (10)

From (9):

$$-\eta_{kk} + \alpha_1 + \alpha_2 \eta_{xk} = 0$$

From (10):

$$-\eta_{xk} + \alpha_4 \eta_{kk} + \alpha_5 \eta_{xk} \eta_{kk} = 0$$

$\longrightarrow$ We want to compute η_{kk}, as it links k_t to k_{t-1}

$\longrightarrow$ Solve for η_{xk} from (9), substitute in (10):

$$\eta_{xk} = -\frac{\alpha_1}{\alpha_2} + \frac{\eta_{kk}}{\alpha_2}$$

$$\Rightarrow + \frac{\alpha_1}{\alpha_2} - \frac{\eta_{kk}}{\alpha_2} + \alpha_4 \eta_{kk}$$

$$- \alpha_5 \frac{\alpha_1}{\alpha_2} \eta_{kk} + \alpha_5 \frac{\eta_{kk}}{\alpha_2} \eta_{kk} = 0$$

or, multiply by $\frac{\alpha_2}{\alpha_5}$ each term,

$$\frac{\alpha_1}{\alpha_5} - n_{kk}\frac{1}{\alpha_5} + n_{kk}\,\alpha_4\,\frac{\alpha_2}{\alpha_5} - \alpha_1 n_{kk} + n_{kk}^2 = 0$$

$$\boxed{0 = n_{kk}^2 - n_{kk}\left(\alpha_1 - \frac{\alpha_2}{\alpha_5}\alpha_4 + \frac{1}{\alpha_5}\right) + \frac{\alpha_1}{\alpha_5}}$$

→ Use the solution to a quadratic equation to find the roots, i.e. $n_{kk,1}$ and $n_{kk,2}$

→ obs: to obtain n_{xk}, go back to

$$n_{xk} = -\frac{\alpha_1}{\alpha_2} + \frac{n_{kk}}{2}$$

the stable one: $|n_{kk}| < 1$

At this point we are done with the coefficients for k_{t-1}.
Now let us calculate the coefficients for z_t

→ We collect its coefficients, one from (9) and one from (10):

$$-n_{tz} + \alpha_2 n_{xz} + \alpha_3 = 0$$

$$-n_{xz} + \alpha_4 n_{tz} + \alpha_5 n_{xt} n_{tz} + (\alpha_5 n_{xz} + \alpha_6)\varphi = 0$$

$\longrightarrow$ we just calculated n_{xt}

$\Longrightarrow$ the unknowns are:

n_{tz} , n_{xz}

$\longrightarrow$ It is a linear system of 2 equations and 2 unknowns, which we can solve

$\longrightarrow$ The result is in the Presentation $\odot$